DBMS Experiential Learning – Level 3 Project

Project Requirements

1. **Project Title**
   - Each team must identify a **unique project title within the section.**
   - The project should address a meaningful real-world problem and should not be a simple CRUD application.
2. **Team Formation**
   - Projects should be completed as 2 member teams as approved by the course instructor.
   - Each team member must have clearly defined responsibilities and contribute to both database and application development.
3. **Database Requirement**
   - The project must demonstrate **practical database design and implementation.**
   - Students has to use:

     Option 1: SQL + NoSQL

     - **SQL Database (Mandatory):** MySQL or PostgreSQL
     - **NoSQL Database:** MongoDB, Neo4j, etc.

     Option 2: SQL + Vector Database

     - **SQL Database (Mandatory):** MySQL or PostgreSQL
     - **Vector Database:** PostgreSQL with pgvector, Qdrant, Pinecone, Chroma, etc.
   - A project may use two **(Minimum) or more database paradigms**, depending on its requirements.
4. **Database Design**
   - For SQL-based components, students must prepare an **ER diagram** and convert it into a **relational schema.**
   - The database design must identify:
     - Entities/tables
     - Attributes
     - Primary keys
     - Foreign keys
     - Relationships
     - Appropriate constraints
   - For NoSQL/vector components, students must document the **data model, collections/documents, indexes, embeddings, metadata, and retrieval strategy**, as applicable.
5. **SQL Implementation**
   - Students using SQL must create and execute appropriate **DDL statements.**

- The project should demonstrate suitable use of:
  - `CREATE`
  - `ALTER`
  - `DROP`
  - Primary and foreign keys
  - Constraints
  - Indexes
  - Views, where appropriate

6. **Database Operations**
   - The application must demonstrate meaningful database operations using **DML**:
     - `INSERT`
     - `UPDATE`
     - `DELETE`
     - `SELECT`
   - Appropriate queries involving **JOIN, GROUP BY, HAVING, ORDER BY, aggregate functions, subqueries, etc.** should be demonstrated wherever applicable.
7. **Advanced Database Features**
   - Each project should demonstrate at least **four advanced database concepts**, such as:
     - Transactions
     - Views
     - Stored procedures/functions
     - Triggers
     - Indexing
     - Full-text search
     - NoSQL aggregation
     - Vector similarity search
8. **Vector Database / Semantic Search — Optional but Recommended**
   - Projects involving unstructured or text-heavy data are encouraged to implement **vector embeddings and similarity search.**
   - Students should explain:
     - What data is converted into embeddings
     - Which embedding approach/model is used
     - How vectors are stored
     - Similarity metric used
     - How relevant results are retrieved
9. **Web Application Interface**
   - The project must provide a functional **web-based interface using Next.js.**
   - The interface should allow users to interact with the implemented database functionality.
   - The application should include appropriate forms, tables, search, filtering, dashboards, or other UI components relevant to the project.

10. **Database–Application Integration**
    - The Next.js application must communicate with the database through an appropriate backend/API layer.
    - Database credentials and sensitive configuration must **not be hard-coded** in the source code.
    - Environment variables should be used for database configuration.
11. **Authentication and Authorization**
    - Projects should implement appropriate **user authentication** where relevant.
    - Different roles such as **Admin, Staff, Student, Customer, etc.** should be considered when the problem domain requires role-based access(Minimum 2 roles)
12. **Data Validation and Integrity**
    - Input validation must be implemented at the database level.
    - The project should demonstrate how **data consistency and integrity** are maintained.
13. **Security**
    - Students must follow basic database and web security practices.
    - Passwords and API/database credentials must not be stored in plain text.
14. **Project Documentation**

    Each team must maintain documentation containing:
    - Problem statement
    - Objectives
    - Scope
    - Functional requirements
    - ER diagram
    - Relational schema
    - Database choice and justification
    - Data dictionary/schema description
    - SQL/NoSQL/vector database implementation
    - Important queries
    - System architecture
    - Application screenshots
    - Testing results
    - Conclusion and future enhancements
15. **Git/GitHub**
    - Each team should maintain the project source code using **Git/GitHub**.
    - Commits should demonstrate **individual contributions** of team members.
    - The repository should contain a suitable `README` with setup and execution instructions.
16. **Evaluation / Demonstration**

    The final demonstration should include:
    - Problem statement and motivation
    - ER diagram and database design
    - Relational schema, where applicable

- Database implementation
- Important queries/operations
- Next.js interface
- Database integration
- Advanced database features
- Test cases
- Individual contribution of team members

17. **Originality**
    - Projects should demonstrate **original design and implementation.**
    - Directly copying an existing project, tutorial, or repository is not permitted.
    - Teams working on similar domains must have substantially different **problem statements, features, database designs, and implementations.**
18. **Level 3 Expectation**
    - The project should go beyond basic **Create–Read–Update–Delete (CRUD)** functionality.
    - Students are expected to demonstrate a clear connection between **database concepts, database design, implementation, and a functional web application.**
    - The choice of SQL, NoSQL, and/or vector database should be **justified based on the nature of the data and application requirements**, rather than being included merely for technology demonstration.

Suggested minimum technology stack

Frontend: Next.js
Backend/API: Next.js API routes / Route Handlers or an appropriate backend service
Database: SQL with NoSQL / Vector Database
Version Control: Git + GitHub

“Database technology must be selected based on the problem requirements, and students must justify why SQL, NoSQL, or a vector database is appropriate for each type of data.”